

Multiple Choice (20 marks)

1. A change in a neuron membrane potential from +50 millivolts to -70 millivolts is considered
 - (a) depolarization
 - (b) repolarization
 - (c) hyperpolarization
 - (d) an action potential
2. An animal cell that is permeable to water but not salts has an internal NaCl concentration of 10%. If placed in freshwater the cell will
 - (a) plasmolyze
 - (b) swell and eventually lyse
 - (c) endocytose water into a large central vacuole
 - (d) shrivel
3. Which of the following processes occur in the cytoplasm of an eukaryotic cell? I. DNA replication II. Transcription III. Translation
 - (a) I only
 - (b) III only
 - (c) II and III only
 - (d) I, II, and III
4. The frequency of a particular allele in a population of 1,000 birds in Hardy-Weinberg equilibrium is 0.3. If the population remains in equilibrium, what...
 - (a) It will increase if the allele is favorable or will decrease if it is unfavorable for individuals in that population.
 - (b) Not enough information is provided to determine if the frequency of the allele will remain the same or change.
 - (c) The frequency of the allele will remain at 0.3 because the population is at Hardy-Weinberg equilibrium.
 - (d) The frequency of the allele will remain the same; this is an example of the bottleneck effect.
5. Hawkmoths are insects that are similar in appearance and behavior to hummingbirds. Which of the following is LEAST valid?

- (a) These organisms are examples of convergent evolution.
 - (b) These organisms were subjected to similar environmental conditions.
 - (c) These organisms are genetically related to each other.
 - (d) These organisms have analogous structures.
6. The normal pH of the cytosol is about 7.2. Most enzymes in the lysosome, however, are only active at a pH of 5. This suggests that
- (a) only acidic substances can undergo enzymatic processing in lysosomes
 - (b) most enzymes in the lysosome are incapable of catalyzing substrates inside the lysosome
 - (c) enzymes in the lysosome are released to the cytosol
 - (d) protons are actively pumped into the lysosome
7. Which of the following must exist in a population before natural selection can act upon that population?
- (a) Genetic variation among individuals
 - (b) Variation among individuals caused by environmental factors
 - (c) Sexual reproduction
 - (d) B and C only
8. Crop rotation is the successive planting of different crops on the same land in sequential seasons. Which of the following statements summarizes the most...
- (a) It allows the farmer to plant the crops with the greatest demand in the marketplace.
 - (b) It increases crop yields and soil fertility and minimizes pathogen infection and pest infestation.
 - (c) It allows the hybridization of successive crops and the development of newer varieties of crop plants.
 - (d) It decreases the need for diverse harvesting equipment and minimizes production costs.
9. Which of the following is not a form of interspecies interaction?
- (a) Commensalism
 - (b) Succession
 - (c) Mutualism
 - (d) Parasitism
10. Invertebrate immune systems possess which of the following?

- (a) Cytotoxic T-lymphocytes
- (b) Phagocytes
- (c) B-cells
- (d) Helper T-cells

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Which of the following is (are) required when performing a polymerase chain reaction (P C R) on a DNA sample?' is 'Reverse transcriptase'.
2. True or False: The correct answer to 'Imagine an organism whose $2n = 96$. Meiosis would leave this organism's cells with how many chromosomes?' is '48'.
3. True or False: The correct answer to 'A healthy man and woman are worried about having children because they are both carriers of sickle-cell disease. If the couple have a...' is '75%'.
4. True or False: The correct answer to 'All of the following are modes of asexual reproduction EXCEPT' is 'meiosis'.
5. True or False: The correct answer to 'This biome has cold winters and is known for its pine forests.' is 'Deciduous forests'.
6. True or False: The correct answer to 'To ensure that hatchling sea turtles orient to the sea rather than inland, which of the following is the most environmentally sound...' is 'Building low fences on the sand dunes'.
7. True or False: The correct answer to 'A new plant was discovered and determined to have an unusually low number of stomata on the undersides of its leaves. For what...' is 'Humid and sunny'.
8. True or False: The correct answer to 'Which of the following disorders will most likely result from X chromosome aneuploidy in women?' is 'Hemophilia'.
9. True or False: The correct answer to 'All of the following statements are true EXCEPT' is 'insulin decreases storage of glycogen'.
10. True or False: The correct answer to 'Which of the following best explains why lignin is important to the evolutionary success of land plants?' is 'Lignin supplies the developing embryo with essential nutrients, enabling its growth.'.

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